

Generalized Robinson Correspondence

Fomin's Growth Diagrams

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May 2026

Partitions and Young diagrams

Definition

A *partition* is a sequence

$$\lambda = (\lambda_1, \lambda_2, \dots)$$

of nonnegative integers in decreasing order:

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots,$$

with only finitely many nonzero terms.

For example, $\lambda = (4, 2, 1)$ is a partition of 7, since $4 + 2 + 1 = 7$. We write $\lambda \vdash 7$.

Young diagrams are the shapes used to represent partitions:

$$(4, 2, 1) \longleftrightarrow \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & & \\ \hline \square & & & \\ \hline \end{array}$$

Definition

A *standard Young tableau* of shape $\lambda \vdash n$ is a filling of the Young diagram of λ with $1, 2, \dots, n$ such that the entries increase along each row and down each column.

Example of a standard Young tableau of shape $(3, 2)$:

1	2	4
3	5	

Definition

A poset P is called a *graded poset* if there exists a rank function $\rho : P \rightarrow \mathbb{Z}_{\geq 0}$ satisfying:

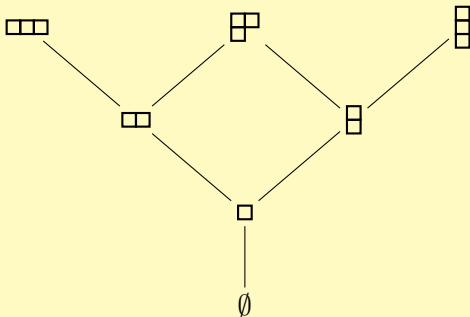
- (i) $\rho(y) = \rho(x) + 1$ whenever $x \lessdot y$ in P , and
- (ii) $\rho(x) = 0$ for all minimal elements $x \in P$.

The value $\rho(x)$ is called the *rank* or *degree* of x .

Young's lattice

Young's lattice $\mathbb{Y}$ is the graded poset whose vertices are partitions.

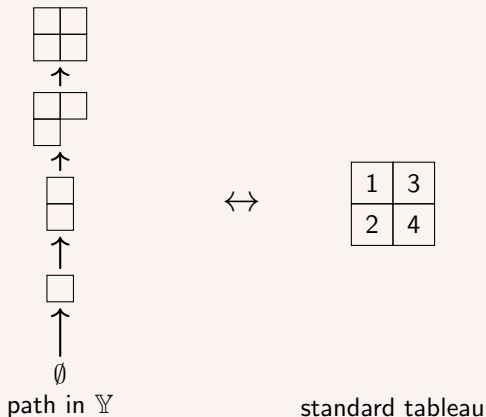
We write $\mu \lessdot \lambda$ if λ is obtained from μ by adding one box.



The grading is the number of boxes.

A path in the Young lattice from $\emptyset$ to a partition λ is equivalent to a standard Young tableau of shape λ .

Example



Young–Fibonacci Lattice

Definition

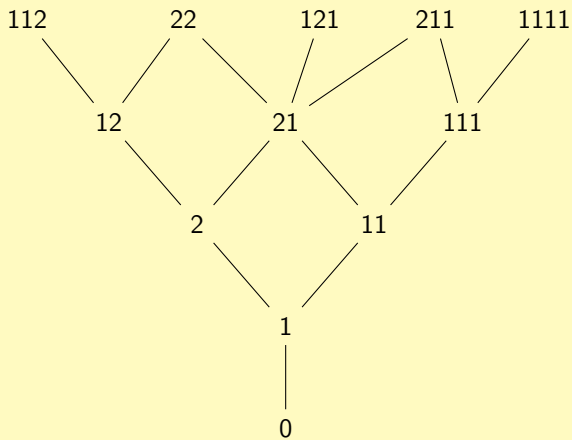
Let V be the set of all finite words in the alphabet $\{1, 2\}$. The *Young–Fibonacci lattice* $\mathbb{YF}$ is the poset with covering relation:

$$s \lessdot w$$

if s is obtained from w by:

- removing the first one, or
- replacing any two standing before the first one, by 1.

Equivalently we get the elements at the n the level (sum of integers is n) recursively as follows; it is the union of elements obtained by prepending 2 to the elements at the $(n - 2)$ th level or prepending 1 to the elements at the $(n - 1)$ th level. Thus the number of elements at the n th level is the n th **Fibonacci number**. .



The Young-Fibonacci lattice is also a graded graph, where the grading is the sum of the digits.

Motivation for Robinson correspondence

For a vertex λ of a graded poset P , define $f_\lambda = \#\{\text{paths from } 0 \text{ to } \lambda\}$.

Example (Young lattice)

Level 3:

$$f_{(3)} = 1, \quad f_{(2,1)} = 2, \quad f_{(1,1,1)} = 1$$

$$\sum_{|\lambda|=3} f_\lambda^2 = 1^2 + 2^2 + 1^2 = 6 = 3!$$

Level 4:

$$f_{(4)} = 1, \quad f_{(3,1)} = 3, \quad f_{(2,2)} = 2, \quad f_{(2,1,1)} = 3, \quad f_{(1,1,1,1)} = 1$$

$$\sum_{|\lambda|=4} f_\lambda^2 = 1^2 + 3^2 + 2^2 + 3^2 + 1^2 = 24 = 4!$$

Example (Young–Fibonacci lattice)

Level 3:

$$f_{12} = 1, \quad f_{21} = 2, \quad f_{111} = 1$$

$$\sum_{|\lambda|=3} f_{\lambda}^2 = 1^2 + 2^2 + 1^2 = 6 = 3!$$

Level 4:

$$f_{1111} = 1, \quad f_{211} = 3, \quad f_{121} = 2, \quad f_{112} = 1, \quad f_{22} = 2$$

$$\sum_{|\lambda|=4} f_{\lambda}^2 = 1^2 + 3^2 + 2^2 + 1^2 + 2^2 = 24 = 4!$$

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Fact

It turns out this is true in general:

$$\sum_{|\lambda|=n} f_{\lambda}^2 = n!$$

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Why does this happen?

There is a bijection

$$S_n$$



pairs of paths (P, Q) from 0
ending at the same vertex at level n

This is the classical **Robinson correspondence**.

Surprisingly, this is not only true for $\mathbb{Y}$ and $\mathbb{YF}$, but for a wide class of posets called **differential posets**.

Skew diagrams and generalized permutations

Definition

Consider $\mathbb{Z}^2$ with partial order

$$(x_1, y_1) \leq (x_2, y_2) \quad \text{if} \quad x_1 \leq x_2, y_1 \leq y_2.$$

A finite convex subset $S \subseteq \mathbb{Z}^2$ is called a *skew diagram*. The *upper/lower boundary* of S :

$$\partial^+(S) = \{(x, y) \in S : (x+1, y+1) \notin S\}, \quad \partial^-(S) = \{(x, y) \in S : (x-1, y-1) \notin S\}.$$

A *generalized permutation* σ is a finite set of cells of S such that no two lie in the same row or column.

A classical example of skew diagram is

$$Sq_n = \{(x, y) \in \mathbb{Z}^2; 0 \leq x \leq n, 0 \leq y \leq n\}.$$

Growths

Definition

Let P and Q be graded posets. A map

$$g : P \rightarrow Q$$

is called a *growth* if

$$x \leq y \implies g(x) \leq g(y).$$

A growth on $\mathbb{Z}^2$ is called a *two-dimensional growth*.

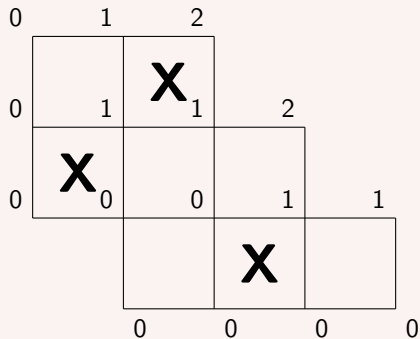
Example

Examples of growths:

1. The rank function $\rho : P \rightarrow \mathbb{Z}$.
2. A multichain in P is a growth from $\mathbb{Z}$ to P .
3. The composition of two growths.

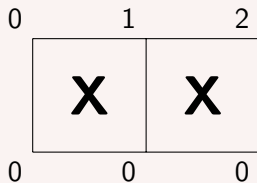
Example

Below is a map $g : S \rightarrow \mathbb{Z}$, where $g(v) = \#$ cells marked with an X which are below and to the left of vertex v . This is a two dimensional growth.



Example

But the following is not a growth.



Definition

A graded poset P is called a *differential poset* if:

1. P is locally finite, graded, and has a unique minimal element 0.
2. For distinct $x, y \in P$, the sets

$$C^+(x) \cap C^+(y) \quad \text{and} \quad C^-(x) \cap C^-(y)$$

are either both empty or both consist of one element where

$$C^+(x) = \{y : x \leq y\}, \quad C^-(x) = \{y : y \leq x\}.$$

3. For every $x \in P$,

$$|C^+(x)| = |C^-(x)| + 1.$$

Lemma

$\mathbb{Y}$ and $\mathbb{Y}F$ are differential posets.

Semimodular growths

We would like to characterize those two-dimensional growths which come from generalized permutations.

Definition

A two-dimensional growth $g : S \rightarrow \mathbb{Z}$ is called *semimodular* if the values of g on the vertices of any cell, say $g_{00}, g_{01}, g_{10}, g_{11}$ satisfy

$$g_{00} + g_{11} \geq g_{01} + g_{10}.$$

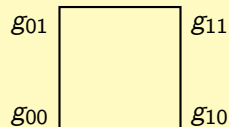
A cell for which the inequality is strict is called an *atom* of g .



The set of atoms of a semimodular growth forms a generalized permutation.

Local extension problem

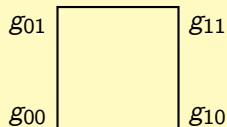
Let $g : S \rightarrow P$ be a growth, where P is a differential poset. Consider the value of g at the four vertices of a cell as in the figure below.



We want to extend the growth from the upper boundary or lower boundary of the skew diagram S to the whole of S .

Local extension problem

Let $g : S \rightarrow P$ be a growth, where P is a differential poset. Consider the value of g at the four vertices of a cell as in the figure below.



We want to extend the growth from the upper boundary or lower boundary of the skew diagram S to the whole of S .

How do we do this?

First we try to extend the growth from the upper boundary of S . If we attempt to compute the value of g_{00} given the other three vertices, we have several cases.

Fomin's Local Growth Rules

Case 1

$$|g_{01}| < |g_{10}|$$

$$g_{00} = g_{01}$$

$$g_{11} = g_{10}$$

Fomin's Local Growth Rules

Case 1

$$|g_{01}| < |g_{10}|$$

$$g_{00} = g_{01}$$

$$g_{11} = g_{10}$$

Case 2

$$|g_{10}| < |g_{01}|$$

$$g_{00} = g_{10}$$

$$g_{11} = g_{01}$$

Fomin's Local Growth Rules

Case 1

$$|g_{01}| < |g_{10}|$$

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Case 2

$$|g_{10}| < |g_{01}|$$

$$g_{00} = g_{10}$$

$$g_{11} = g_{01}$$

Case 3

$$|g_{01}| = |g_{10}|$$

$$g_{01} \neq g_{10}$$

$$g_{00} = g_{01} \cap g_{10}$$

$$g_{11} = g_{01} \cup g_{10}$$

These are the forced cases.

Case 4

Let $g_{01} = g_{10} := x \in P$. Since P is differential,

$$|C^+(x)| = |C^-(x)| + 1.$$

Thus we choose a bijection

$$\phi_x : C^+(x) \longrightarrow C^-(x) \cup \{x\}.$$

We also use the convention

$$\phi_x(x) = x.$$

Then, when extending downward, define

$$g_{00} = \phi_x(g_{11}).$$

Note that the growth we obtained thus on S is in fact semimodular.

Extending Upwards

Now suppose g_{00}, g_{01}, g_{10} are known and we want to determine g_{11} .

The first three cases are forced as before.

Case 4

The ambiguity occurs when $g_{00} = g_{01} = g_{10} = x$. This time we need the help of a generalized permutation on the cells of S .

- If $g_{00} \neq x$, set

$$g_{11} = \phi_x^{-1}(g_{00}).$$

- If $g_{00} = x$ and the cell is not marked with an X , set

$$g_{11} = x.$$

- If $g_{00} = x$ and the cell is marked by X , choose the element $y \in C^+(x)$ such that $\phi_x(y) = x$, and set

$$g_{11} = y.$$

Definition

An *R-correspondence* on a differential poset P is a collection $\Phi = \{\phi_x\}_{x \in P}$ of bijections $\phi_x : C^+(x) \longrightarrow C^-(x) \cup \{x\}$. A growth $g : S \rightarrow P$ is called a Φ -*growth* if whenever $g_{01} = g_{10} = x$, we have $g_{00} = \phi_x(g_{11})$.

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Lemma

Let P be a differential poset and S be a skew diagram. Any growth $g^+ : \partial^+(S) \rightarrow P$ can be uniquely extended to a Φ growth g on all of S . Furthermore $|g|$ will be semimodular.

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Lemma

Let P be a differential poset, S a skew diagram and σ a generalized permutation on the cells of S . Then any growth $f^- : \partial^-(S) \rightarrow P$ can be extended uniquely to a Φ growth g on all of S in such a way that $|g|$ is semimodular with the atoms of g being the cells in σ .

Fomin's Theorem

Theorem (Fomin)

Let P be a differential poset, S a skew diagram, and Φ an R -correspondence. Then there is a bijection

$$\{ \text{growths } g^+ : \partial^+(S) \rightarrow P \} \longleftrightarrow \left\{ \begin{array}{l} (g^- : \partial^-(S) \rightarrow P, \sigma) \\ \text{where } \sigma \text{ is a generalized permutation} \end{array} \right\}.$$

Each of these is uniquely associated with a Φ -growth $g : S \rightarrow P$, whose restriction to upper and lower boundary are g^+ and g^- respectively, and whose modulus $|g|$ is semimodular growth whose atoms are σ .

Scheme of the Generalized Robinson Correspondence

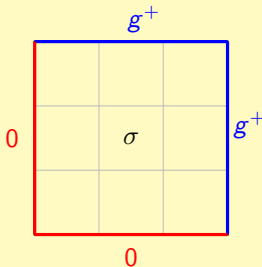
Let $S = Sq_n$ be the square grid.

For a differential poset P , a pair of paths

$$P_1, P_2 : 0 \longrightarrow \lambda, \quad |\lambda| = n,$$

can be viewed as a growth on the upper boundary:

$$g^+ : \partial^+(Sq_n) \rightarrow P.$$



Fix the lower boundary growth of Sq_n to be constantly 0. The upper boundary growth is strict if and only if the corresponding generalized permutation is a standard permutation in S_n .

Classical Robinson Correspondence

By Fomin's theorem, we have a bijection

$$S_n \longleftrightarrow \{(P, Q) : P, Q \text{ paths from } 0 \text{ ending at the same vertex at level } n\}.$$

Proposition

If we have the following bijection by Robinson correspondence;

$$\pi \xleftrightarrow{R} (P, Q) \quad \text{then} \quad \pi^{-1} \xleftrightarrow{R} (Q, P).$$

Fomin's Local Growth Rules on Young Lattice

To specialize Fomin's theorem to the Young lattice $\mathbb{Y}$, we first fix an R -correspondence.

R -correspondence on $\mathbb{Y}$

Let μ be a partition and let $\lambda \in C^+(\mu)$. Suppose λ is obtained from μ by adding one box in the i -th row.

Define $\phi_\mu(\lambda)$ by:

- if $i = 1$, then $\phi_\mu(\lambda) = \mu$;
- if $i > 1$, then $\phi_\mu(\lambda)$ is obtained from μ by deleting the last box of the $(i - 1)$ -st row.

Fix

$$\mu = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \\ \hline \square & & \\ \hline \end{array}$$

Example 1

$$\phi_{\mu} \left(\begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \square & \square & & \\ \hline \square & & & \\ \hline \end{array} \right) = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \\ \hline \square & & \\ \hline \end{array}$$

Example 2

$$\phi_{\mu} \left(\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \square & & \\ \hline \end{array} \right) = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \\ \hline \end{array}$$

Example

For the permutation:

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 5 & 6 & 4 & 3 \end{pmatrix},$$

we have the following growth diagram:

Local Rule on Vertices

$\emptyset$	1	X 2	21	22	32	321	3211
$\emptyset$	1	1	11	21	X 31	32	321
$\emptyset$	1	1	11	X 21	21	22	221
$\emptyset$	1	1	11	11	11	X 21	22
$\emptyset$	1	1	11	11	11	11	X 21
$\emptyset$	X 1	1	11	11	11	11	11
$\emptyset$	$\emptyset$	$\emptyset$	X 1	1	1	1	1
$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$	$\emptyset$

If we look at the vertices on the top part of the upper boundary or the right part of the upper boundary of Sq_n , we see that the corresponding Young diagrams represent a pair of paths in $\mathbb{Y}$ which equivalently determine a pair of standard tableaux in an obvious manner. The right boundary determine the standard tableau Q and the top boundary determine the standard tableau P :

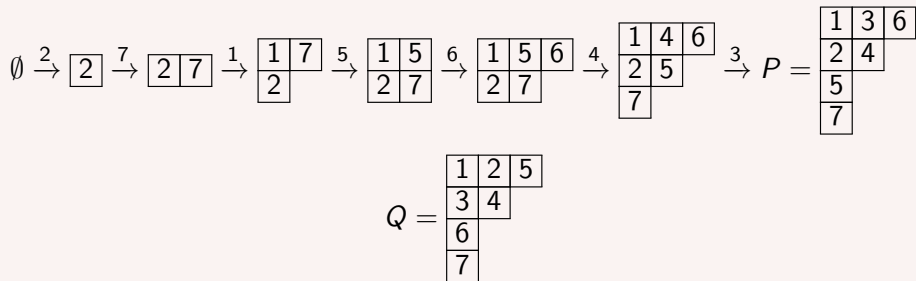
$$P = \begin{array}{|c|c|c|} \hline 1 & 3 & 6 \\ \hline 2 & 4 & \\ \hline 5 & & \\ \hline 7 & & \\ \hline \end{array}, \quad Q = \begin{array}{|c|c|c|} \hline 1 & 2 & 5 \\ \hline 3 & 4 & \\ \hline 6 & & \\ \hline 7 & & \\ \hline \end{array}$$

Insertion Algorithm (Row Bumping)

Example

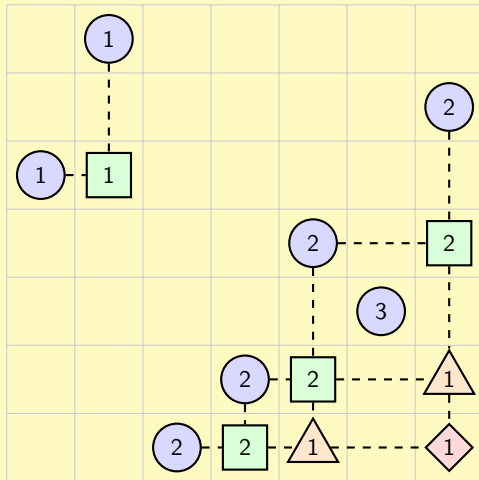
Consider the permutation

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 5 & 6 & 4 & 3 \end{pmatrix}.$$



Matrix Ball Construction

Consider the same permutation $\pi \in S_7$ as before. The process of matrix ball construction is schematically shown as:



Global Rule on Vertices

Given a finite poset we want to associate a Young diagram to it. For any vertex v of Sq_n , define $C(v)$ to be the set of cells of S below and left to v . Also note that the set of cells of Sq_n form a poset in an obvious manner. Let σ be a generalized permutation and for a poset P , $\lambda(P)$ denote the Young diagram corresponding to the poset obtained using Greene's theorem.

Theorem (Greene)

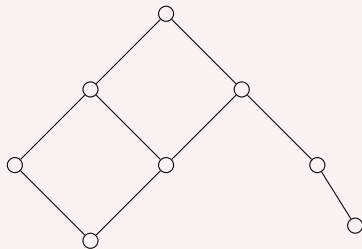
Let P be a finite poset. Set $\lambda_k(P) = c_k(P) - c_{k-1}(P)$ and $\mu_k(P) = a_k(P) - a_{k-1}(P)$, where $c_k(P)$ (resp. $a_k(P)$) is the size of the longest union of k chains (resp. antichains). Then $\lambda(P)$ and $\mu(P)$ are partitions, and μ is the conjugate of λ .

Define a growth $G : Sq_7 \rightarrow \mathbb{Y}$ (where $\mathbb{Y}$ is the Young lattice)

$$v \mapsto \lambda(C(v) \cap \sigma).$$

Example

Consider the following poset.

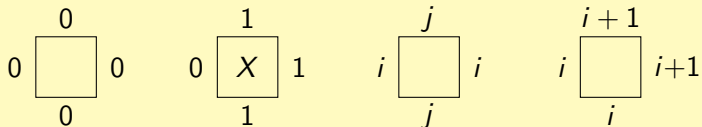


Corresponding to the above theorem, the values of $c_k(P)$ are $c_0 = 0$, $c_1 = 4$, $c_2 = 7$, $c_3 = 8$, $c_4 = 8, \dots$ thus the Young diagram corresponding to this poset has shape $(4, 3, 1)$.

The growth diagram defined by G yields the same growth diagram we obtained by Fomin's local rule on vertices.

Local Rule on Edges

The rule can be described by the following diagrams.



In the third case assume $i \neq j$.

Example

Once again, consider the permutation:

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 7 & 1 & 5 & 6 & 4 & 3 \end{pmatrix}.$$

We obtain the following diagram with labeling:

	1	1	2	2	1	3	4
0	0	X 1	1	2	2	3	4
0	1	0	2	1	1	2	3
	0	0	0	0	X 1	1	1
0	1	0	2	1	0	2	3
	0	0	0	X 1	1	2	3
0	1	0	2	0	0	1	2
	0	0	0	0	0	X 0	2
0	1	0	2	0	0	0	1
	0	0	0	0	0	0	X 1
0	1	0	2	0	0	0	0
	X 1	1	2	2	2	2	2
0	0	0	1	0	0	0	0
	0	0	X 1	1	1	1	1
	0	0	0	0	0	0	0

Fomin's Local Growth Rules on Young-Fibonacci Lattice

To apply Fomin's theorem to $\mathbb{YF}$, we first fix an R -correspondence.

R -correspondence on $\mathbb{YF}$

Let $x \in \mathbb{YF}$. The elements covering x are of the form $1x$ and $2z$, where z lies two levels below x . Define

$$\phi_x(1x) = x, \quad \phi_x(2z) = z.$$

Thus ϕ_x sends an element covering x either to x itself or to an element covered by x .

Example

Let

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 9 & 1 & 8 & 6 & 7 & 3 \end{pmatrix}.$$

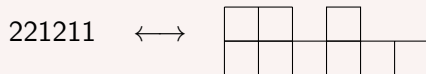
Then applying the local rules on vertices we obtain the following growth diagram:

0	1	11	21	¹²¹ X	221	2211	21211	22211	221211
0	1	11	21	21	211	¹²¹¹ X	2211	21211	211211
0	1	11	21	21	211	211	1211	¹¹²¹¹ X	21211
0	1	11	21	21	211	211	¹²¹¹ X	1211	2211
0	1	¹¹ X	21	21	211	211	211	211	221
0	1	1	¹¹ X	11	21	21	21	21	22
0	1	1	1	1	2	2	2	2	¹² X
0	¹ X	1	1	1	2	2	2	2	2
0	0	0	0	0	¹ X	1	1	1	1
0	0	0	0	0	0	0	0	0	0

Representing words and paths in $\mathbb{Y}F$

We represent an element of $\mathbb{Y}F$ typographically as follows: each 1 corresponds to one box, 2 corresponds to two boxes stacked vertically, and we read the word from left to right.

Example



For the pair of paths obtained from Fomin's local rules, we can try to represent them by tableaux:

$$P = \begin{array}{|c|c|c|c|c|c|} \hline 4 & 9 & & 2 & & \\ \hline 3 & 8 & 7 & 1 & 6 & 5 \\ \hline \end{array} \quad Q = \begin{array}{|c|c|c|c|c|c|} \hline 5 & 8 & & 3 & & \\ \hline 4 & 7 & 9 & 2 & 6 & 1 \\ \hline \end{array}$$

These shapes, corresponding to paths in $\mathbb{Y}F$, are called *path tableaux*.

Why path tableaux are not enough

Path tableaux are less convenient than standard Young tableaux. Even if each entry in the top row is greater than the one below it, the filling need not come from a path in $\mathbb{Y}F$.

Example

Let

$$T = \begin{array}{|c|c|c|c|} \hline 6 & 2 & & 7 \\ \hline 5 & 1 & 3 & 4 \\ \hline \end{array}$$

Then it should represent the path

$$P = (0, 1, 2, 21, 211, 1211, 2211, 2212).$$

But this does not represent a path, since

$$2212 \text{ does not cover } 2211 \text{ in } \mathbb{Y}F.$$

Insertion algorithm for $\mathbb{Y}F$

Suppose we want to insert a number a into a partial tableau T .

- Compare a with the leftmost entry t_1 in the bottom row.
- If $a > t_1$, attach a new cell to the left of t_1 , put a there, and stop.
- If $a < t_1$, move a into the cell immediately above t_1 , bumping the entry there.
- Continue inductively.

Example

Inserting 8 into

1	4	
9	5	2

gives

8	1		
9	5	4	2

Example

Consider

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 5 & 4 & 9 & 1 & 8 & 6 & 7 & 3 \end{pmatrix}.$$

Inserting the second row entries gives:

$$\begin{array}{|c|}, & \begin{array}{|c|c|}\hline 5 & 2 \\\hline\end{array}, & \begin{array}{|c|c|}\hline 4 & \\ \hline 5 & 2 \\\hline\end{array}, & \begin{array}{|c|c|c|}\hline & 4 & \\ \hline 9 & 5 & 2 \\\hline\end{array}, & \begin{array}{|c|c|c|}\hline 1 & 4 & \\ \hline 9 & 5 & 2 \\\hline\end{array}, & \begin{array}{|c|c|c|c|c|}\hline 8 & 1 & & & \\ \hline 9 & 5 & 4 & 2 & \\ \hline\end{array}\end{array}$$

$$\begin{array}{|c|c|c|c|c|}\hline 6 & & 1 & & \\ \hline 9 & 8 & 5 & 4 & 2 \\\hline\end{array}, \quad \begin{array}{|c|c|c|c|c|}\hline 7 & 6 & 1 & & \\ \hline 9 & 8 & 5 & 4 & 2 \\\hline\end{array}, \quad \begin{array}{|c|c|c|c|c|}\hline 3 & 7 & & 1 & \\ \hline 9 & 8 & 6 & 5 & 4 & 2 \\\hline\end{array}$$

$$P = \begin{array}{|c|c|c|c|c|c|}\hline 3 & 7 & & 1 & & \\ \hline 9 & 8 & 6 & 5 & 4 & 2 \\\hline\end{array}$$

$$\text{recording tableau } Q = \begin{array}{|c|c|c|c|c|c|}\hline 5 & 8 & & 3 & & \\ \hline 4 & 7 & 9 & 2 & 6 & 1 \\\hline\end{array}$$

Tableaux for $\mathbb{Y}F$

Now there is no obvious way to retrieve the path from the tableau P .

Definition

The shapes with fillings such that entries decrease along the bottom row and up each column are called *tableaux*.

For $x \in \mathbb{Y}F$, define $ZT(x) = \{\text{tableaux of shape } x\}$, and $PZT(x) = \{\text{path tableaux of shape } x\}$. Set $z_x := |ZT(x)|$.

Proposition

There is a bijection between permutations $\pi \in S_n$ and pairs (P, Q) such that

$$P \in ZT(x), \quad Q \in PZT(x), \quad |x| = n.$$

We write this as

$$\pi \xleftrightarrow{F-R-S} (P, Q).$$

Theorem

If $\pi \xleftrightarrow{F-R-S} (P, Q)$, then $\pi^{-1} \xleftrightarrow{F-R-S} (Q, P)$.

The theorem is clear when P and Q are considered as paths. The theorem states that any path in $\mathbb{Y}F$ from 0 to x can be written in either of the form; as an element of $PZT(x)$ or $ZT(x)$. For example, Let $P \in ZT(x)$ and we want to view it as an element of $PZT(x)$.

Choose $Q \in PZT(x)$.
Work backwards to recover the permutation π corresponding to (P, Q) .



Apply the algorithm to π^{-1} .
We obtain a pair (A, B) , where $B \in PZT(x)$ represents P in the required form.

Corollary

$$\sum_{|x|=n} z_x^2 = n! \text{ and } \sum_{|x|=n} z_x = \#\{\text{involutions in } S_n\}.$$

Example

Consider the involution

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 2 & 1 & 7 & 6 & 5 & 4 \end{pmatrix}.$$

Then applying insertion algorithm we obtain the tableau:

$$P = \begin{array}{|c|c|c|} \hline 4 & 5 & 1 \\ \hline 7 & 6 & 3 & 2 \\ \hline \end{array}$$

Observe that the cycle decomposition for π is

$$\pi = (13)(2)(47)(56).$$

Theorem

Let $\pi \in S_n$ be an involution and $\pi \xleftrightarrow{F-R-S} P \in ZT(x)$. Then P represents the cycle decomposition of π in an obvious manner.

Global Rule for Young Fibonacci Lattice

Let π be a generalized permutation of the cells of Sq_n . For each vertex v of Sq_n , we define a word $G(v) \in \mathbb{YF}$.

Recursive rule

Start with the empty word 0. Look at the marked cells of π lying below and to the left of v . At each step let R be the marked cell in the highest row, and let C be the marked cell in the highest column,

- if $R = C$, append 1,
- if $R \neq C$, append 2,
- delete R and C , and repeat.

When no marked cells remain, the resulting word is $G(v)$.

Example

Consider the permutation

$$\pi = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}.$$

Then the global algorithm produces the following diagram:

0		1		11		111		211
					X			
0		1		11		11		21
			X					
0		1		1		1		2
	X							
0		0		0		0		1
							X	
0								
		0		0		0		0

References

-  T. Roby, *Applications and Extensions of Fomin's Generalizations of the Robinson-Schensted Correspondence to Differential Posets*, Ph.D. Thesis, Massachusetts Institute of Technology, 1991.
-  W. Fulton, *Young Tableaux: With Applications to Representation Theory and Geometry*, London Mathematical Society Student Texts **35**, Cambridge University Press, Cambridge, 1997.
-  S. Fomin, *Generalized Robinson–Schensted–Knuth Correspondence*, Zapiski Nauchnykh Seminarov LOMI **155** (1986), 156–175; English translation: Journal of Soviet Mathematics **41** (1988), no. 2, 979–991.
-  X. G. Viennot, *The Art of Bijective Combinatorics, Part III: The Robinson–Schensted–Knuth correspondence*, IMSc, Chennai, 2018.
<http://www.viennot.org/abjc2.html>